Yinchang Ma

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Research profile

Functional two-dimensional materials and nanoelectronic devices, with a focus on ferroelectrics and antiferroelectrics, van der Waals magnetism, spintronics, nonvolatile memories, and neuromorphic computing.

Google Scholar snapshot, 14 October 2026: 2,843 citations; h-index 30; i10-index 48.

Appointment and education

July 2025 - Present | Postdoctoral Fellow
KAUST

2021 - 2025 | Ph.D., Materials Science and Engineering
KAUST. Advisor: Professor Xixiang Zhang

2019 - 2021 | M.S., Materials Science and Engineering
KAUST

2014 - 2018 | B.Eng., Microelectronics Science and Engineering
University of Electronic Science and Technology of China, Chengdu

Teaching and mentoring

Guest lecturer for graduate courses *Electronic Properties of Materials* and *Crystallography and Diffraction* at KAUST. Contributions include English-language teaching, laboratory guidance, examination preparation, and student support.

Mentoring Ph.D., M.S., and visiting students in experimental design and data analysis; coordinating international collaborations and contributing to research proposal development.

Research expertise

Materials design & synthesis
AI-assisted density functional theory and high-throughput screening; chemical vapor transport and chemical vapor deposition; composition and defect engineering.

Nanofabrication & integration
L-edit and AutoCAD layout design; electron-beam lithography, laser direct writing, photolithography, sputtering, and reactive-ion etching; van der Waals heterostructure integration.

Electrical & magnetic measurements
Temperature-dependent electrical transport, Hall effect, magnetoresistance, tunneling magnetoresistance, and quantum oscillations; PPMS and SQUID measurements.

Structural & functional characterization
Second-harmonic generation, Raman spectroscopy, SEM, TEM, and piezoresponse force microscopy; semiconductor carrier and mobility analysis.

Research projects

Pushing Orbitronics to the 2D limit
2025 - Present | Participating researcher
KAUST Competitive Research Grants · ORA-CRG12-2025-6036

Spintronics based on novel antiferromagnetic materials with spin-momentum locking
2023 - Present | Participating researcher
KAUST Competitive Research Grants · ORA-CRG11-2023-5031

Direct observation of antiferromagnetic spins in novel spintronic nanostructures
2022 - 2025 | Participating researcher
KAUST Competitive Research Grants · ORA-CRG10-2021-4665

Academic service and recognition

Peer reviewer: Nature Communications; InfoMat; npj Computational Materials; ACS Applied Materials & Interfaces; Materials Science and Engineering: R: Reports; Small Methods; Materials Today Physics; Materials Today Quantum; Journal of the American Ceramic Society; Journal of Alloys and Compounds; Ceramics International; Nanoscale Advances; Microelectronic Engineering; Journal of Superconductivity and Novel Magnetism; Journal of Inorganic and Organometallic Polymers and Materials; Advanced Functional Materials.

KAUST Dean's Award (academic achievement in the top 10%, as recorded in the supplied CV).

Invited presentations at Zhejiang University, UESTC, Nanchang University, and Oxford Instruments forums.

Representative publications

Only first-author, co-first-author, or corresponding-author contributions are included. Author roles follow the supplied CV and available author notes. Full author lists and current citations are available through the DOI and Google Scholar records.

1. Hysteresis Inversion Driven by Polar-Order Competition in van der Waals Ferrielectric

Advanced Materials (2026). First author.
DOI: 10.1002/adma.74102

2. Phase engineering of relaxor ferroelectricity in van der Waals crystal

Nature Communications (2026). Co-first author.
DOI: 10.1038/s41467-026-69272-9

3. Van der Waals Engineering of One-Transistor-One-Ferroelectric-Memristor Architecture for an Energy-Efficient Neuromorphic Array

Nano Letters (2025). First author.
DOI: 10.1021/acs.nanolett.4c06118

4. The van der Waals antiferromagnetic proximity effect at the FePS3/Pt uncompensated interface

Materials Horizons (2025). First author · equal contribution.
DOI: 10.1039/d5mh00023h

5. Machine learning for 2D material-based devices

Materials Science and Engineering: R: Reports (2025). Co-first & co-corresponding author.
DOI: 10.1016/j.mser.2025.101085

6. Identification of the intrinsic, topological, and extrinsic anomalous Hall effects in noncollinear antiferromagnets

Newton (2025). Co-first author.
DOI: 10.1016/j.newton.2025.100055

7. Highly sensitive, responsive, and selective iodine gas sensor fabricated using AgI-functionalized graphene

Nature Communications (2025). Co-first author.
DOI: 10.1038/s41467-025-56621-3

8. Multiscale understanding the effect of K/Na ratio on electrical properties of high-performance KNN-based ceramics

Journal of the American Ceramic Society (2024). Co-corresponding author.
DOI: 10.1111/jace.19465

9. High-performance van der Waals antiferroelectric CuCrP2S6-based memristors

Nature Communications (2023). First author · equal contribution.
DOI: 10.1038/s41467-023-43628-x

10. Two-dimensional ferroelectricity and antiferroelectricity for next-generation computing paradigms

Matter (2022). Co-first author.
DOI: 10.1016/j.matt.2022.05.021

11. Unraveling the origin of ferroelectric resistance switching through the interfacial engineering of layered ferroelectric-metal junctions

Nature Communications (2021). Co-first author.
DOI: 10.1038/s41467-021-27617-6